

Requirements Document: AI-Enabled Skills and Contribution Assessor (POC Phase)

Project Overview

This project aims to create an AI-enabled system leveraging the GitHub Issue API and ChatGPT to summarize intern contributions and classify skills gained per issue in the Hack for LA organization. It generates resume-ready bullet points and adds skill set labels from a predefined CSV taxonomy.

Functional Requirements (POC Scope)

User Input and Interaction

- A predefined list of contributors will be used (no text entry or user input in the POC).

Data Processing

- Fetch comments per issue for each user in the predefined list via GitHub Issue API.
- Extract context from issue overview and action items.
- Submit combined issue context and user comments to ChatGPT.

Output Generation

- **Bullet Points:**
 - Formal style with software development concepts.
 - Use the STAR (Situation, Task, Action, Result) method for clarity.
 - Maximum of 30 words per bullet point.
 - Explicitly state task status (for POC, all issues are closed).
 - Resume bullet comment will follow a standardized format:
 - Opening message describing the purpose, with a link to more detail about the automation.
 - Bullet points grouped per user, e.g.:

```
@username1
- Resume Bullet 1
- Resume Bullet 2

@username2
- Resume Bullet 1
- Resume Bullet 2
```
- **Skill Classification:**

- Based on provided CSV file (HfLA_skills.csv), structured as comma-separated labels.
- Multiple skill labels will be assigned per user.
- Label comment formatting:
 - Opening message
 - Markdown checklist of suggested skills:
 - [] Python
 - [] Project Management
- Ensure API response times and limits from GitHub and OpenAI are manageable.

Non-Functional Requirements

Performance

- The entire operation per issue per user must complete within 1 minute.

Reliability and Error Handling

- Errors from GitHub or OpenAI APIs:
 - Publicly post the error code and user-friendly message to the original issue.
 - Log error details to a structured error log file.
 - These error logs will help identify persistent or pattern-based failures.

Security

- No encryption or anonymization required.
- No anticipated privacy concerns.
- No authentication?

Deployment and Environment

- No existing deployment infrastructure.
- Leveraged existing true-github-contributors package, modifying it for AI-enabled summarization and skill classification tasks.
- Runs daily across all open issues in the Hack for LA organization using a time-triggered script.
- GitHub is used as the database; output files will be written in **CSV and JSON** formats under the /Data folder.
- Schema for these files is TBD but will support:
 - Contributor ID and name
 - Associated issue ID
 - Resume bullets
 - Skill labels

Scalability

- Designed for low concurrent load via sequential batch processing.

Usability

- No GUI; automated backend process using scripts.

Technical Stack and System Notes

- The original JavaScript codebase (true-github-contributors) is being rewritten in **Python**, and all development will use Python going forward.

Data and API Requirements

GitHub API

- Uses GitHub Issue API to fetch comments and post results.
- Requires a bot account with proper collaborator permissions (avoids spam restrictions).
- Uses a **Personal Access Token** (configured to never expire) for authentication.
- Related documentation:
 - [Hack for LA Website Wiki](#)
 - HfLA-Website-Admin: How to update Hackforlabot token

OpenAI API

- GPT-4 Turbo or GPT-4o will be used. Premium ChatGPT API key to be acquired.
- OpenAI rate limits (Tier 1):
 - **RPM (Requests per Minute):** 500
 - **TPM (Tokens per Minute):** 30,000
 - These limits will be respected using throttling logic.
- Excessive token use will be avoided through input size control.
- Consider Costs, per model

Audit and Maintenance

- Log all bot activity, including:
 - User mentions
 - Generated content
 - Errors and diagnostics
- Rotate access tokens periodically.
- Monitor GitHub and OpenAI dashboards for outages.

Future Considerations (Post-POC / Pilot Phase)

- Accepting manual contributor input via UI or CLI

- Handling ongoing (open) issues
 - More granular user notification (e.g., DM vs. group mention)
 - Schema definition and persistence via database or versioned storage
 - Analytics for skill acquisition trends across org
 - Codebase will reside in GitHub repo: [hackforla/ai-skills-assessor](https://github.com/hackforla/ai-skills-assessor).
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This revised document defines the POC phase of the AI Skills Assessor project. It captures constraints, goals, and structural considerations needed to implement, evaluate, and shape a full-scale pilot.